

Project 2 — 自我評估報告

AI Workspace Agent Suite : Calendar Agent + Refund Agent

報告日期：2026-06-02

一、Calendar Agent 測試結果 (對照 PDF 第 1 節)

1.0 測試前置：建立 Table 1 行事曆事件 (§1.2 / §1.3)

由測試注入腳本 `testdata/create_calendar_events.py` (對應 PDF 的 `createcalendarevents.py`) 透過 Google Calendar 自動建立下列 10 筆事件,agent 查詢時確實回傳全部 10 筆,確認注入成功：

日期	時間	事件
Mon Jun 1	09:00–10:00	Team Standup
Mon Jun 1	14:00–15:00	Research Meeting
Tue Jun 2	10:00–11:00	Project Review
Tue Jun 2	15:00–16:00	Student Advising
Wed Jun 3	09:00–10:30	Faculty Meeting
Wed Jun 3	14:00–15:00	PhD Progress Review
Thu Jun 4	11:00–12:00	Industry Collaboration Meeting
Thu Jun 4	15:00–16:00	Lab Weekly Meeting
Fri Jun 5	09:00–10:00	Grant Proposal Discussion
Fri Jun 5	15:00–16:00	Research Seminar

1.1 Test Prompt 1 (§1.4) — What's on my calendar?

對照項目	結果
Expected：取得所有 upcoming events	✓ 呼叫 <code>get_events</code> (範圍 2026-06-01 ~ 06-08),回傳 10 筆事件
Expected：依日期/時間彙整	✓ 輸出依 Mon–Fri 分組,每筆標出時間與標題
Expected：不建立/不修改任何事件	✓ 僅讀取,無任何寫入工具呼叫

對照項目	結果
判定	PASS

ReAct 軌跡：🗨 Thought → ⚙ get_events → 👁 Observation (10 events) → ✅ Final Answer

Agent 實際輸出 (節錄)：

```
📅 Monday, June 1
- 9:00 AM - 10:00 AM - Team Standup
- 2:00 PM - 3:00 PM - Research Meeting
📅 Tuesday, June 2
- 10:00 AM - 11:00 AM - Project Review
- 3:00 PM - 4:00 PM - Student Advising
... (共 10 筆,週末標示「No events」)
You've got 10 events across the weekdays.
```

與測試須知 §1.4 的「Expected Output Example」格式一致 (每筆 日期：事件 (起-迄)) 。

1.2 Test Prompt 2 (§1.5) — Schedule a team lunch for the coming Friday at noon for 1 hour.

對照項目	結果
Expected：檢查行事曆可用性	✅ Thought 中先確認週五既有事件 (9-10、15-16) ,中午無衝突
Expected：建立事件 Title=Team Lunch、Fri 12:00-13:00、1 小時	✅ 呼叫 <code>manage_event(action=create)</code> ,時間 2026-06-05 12:00-13:00 (+08:00)
Expected：確認建立成功	✅ Observation 回傳「Successfully created event 'Team Lunch'」+ 事件連結
判定	PASS (無排程衝突)

ReAct 軌跡：🗨 Thought → ⚙ `manage_event(create)` → 👁 Observation (created) → ✅ Final Answer

Agent 實際輸出 (節錄)：

```
Done! ✅ I've scheduled a Team Lunch for Friday, June 5 at 12:00 PM - 1:00 PM.
Friday now: Grant Proposal Discussion (9-10) / 🍕 Team Lunch (12-1, new) / Research Seminar
```

與 §1.5 「Expected Result : A new event appears on Friday — Team Lunch (12:00–13:00)」完全一致。

1.3 Test Prompt 3 (§1.6) — Find a free 30-minute slot for a call with john@example.com this week.

對照項目	結果
Expected：讀取既有行事曆	✔ 呼叫 query_freebusy (2026-06-01 ~ 06-06),取得 11 段忙碌時段
Expected：找出可用 30 分鐘空檔	✔ 接著對週一~週五各呼叫一次 find_free_slots (自寫工具) 算出空檔
Expected：提出一個以上候選	✔ 提供多個區間,並給出 3 個 top picks
判定	PASS (自由時間正確)

ReAct 軌跡 (兩步推理,最能展現 ReAct)：Thought → query_freebusy → Observation (11 busy) → Thought → find_free_slots x5 → Observation → Final Answer

空檔計算結果 (每日 09:00–18:00, Asia/Taipei)：

日期	計算出的空檔
Mon Jun 1	10:00–14:00、15:00–18:00
Tue Jun 2	09:00–10:00、11:00–15:00、16:00–18:00
Wed Jun 3	10:30–14:00、15:00–18:00
Thu Jun 4	09:00–11:00、12:00–15:00、16:00–18:00
Fri Jun 5	10:00–12:00、13:00–15:00、16:00–18:00

測試須知列出的「Possible Valid Answers」(Monday 10:00–10:30、Tuesday 11:00–11:30、Thursday 13:00–13:30 等) 皆落在上述空檔區間內 → 答案正確。註：john@example.com 為佔位地址,其真實行事曆無法存取 (§1.6 註明「If integrated...」為選用), 故以使用者本人事務曆為準計算空檔。

1.4 Calendar Agent — Success Criteria (§1.7)

Success Criteria	判定
Correct retrieval of existing events	✔ PASS (10/10 筆)

Success Criteria	判定
Correct event creation	✅ PASS (Team Lunch 正確建立)
No scheduling conflicts	✅ PASS (中午時段無衝突)
Accurate free-time discovery	✅ PASS (空檔計算正確)

二、Refund Agent 測試結果 (對照 PDF 第 2 節)

2.0 測試資料 (§2.2 / §2.3)

由測試注入腳本 `testdata/send_test_emails_smtp.py` (對應 PDF 的 `sendtestemails.py` , 從外部 Gmail 帳號經 SMTP 寄入 agent 信箱) 寄出 8 封測試信到受監控信箱,符合規格的 4 類各 2 封 :

類別	主旨
REFUND_REQUEST	Refund Request for Order #1001 / #1002
RETURN_REQUEST	Return Request for Wireless Mouse / Keyboard
COMPLAINT	Very Disappointed / Poor Service Experience
OTHER	Special Summer Promotion / Question About Refund Policy

2.1 處理流程 (§2.4 — `Process all refund emails`)

ReAct 軌跡 (6 步工作流 : SEARCH → READ → CLASSIFY → DRAFT → SEND → REPORT) :

1. **SEARCH** — 三段查詢 (`refund OR return` 、 `complaint OR ...` 、 catch-all `is:unread`) 併合去重 → 8 封唯一信件
2. **READ** — `get_gmail_messages_content_batch` 一次讀取 8 封全文
3. **CLASSIFY** — 依內容分四類 (見下表)
4. **DRAFT / SEND** — 對 6 封可處理信件以對應模板撰寫並 `send_gmail_message` (附 `thread_id` 串接原對話)
5. **REPORT** — 輸出摘要

2.2 逐封分類與處理結果

#	主旨	分類	動作
1	Refund Request for Order #1001	REFUND_REQUEST	✅ Replied (退款核准模板)
2	Refund Request for Order #1002	REFUND_REQUEST	✅ Replied (退款核准模板)
3	Return Request for Wireless Mouse	RETURN_REQUEST	✅ Replied (退貨步驟模板)

#	主旨	分類	動作
4	Return Request for Keyboard	RETURN_REQUEST	✅ Replied (退貨步驟模板)
5	Very Disappointed	COMPLAINT	✅ Replied (致歉 + 24h 跟進)
6	Poor Service Experience	COMPLAINT	✅ Replied (致歉 + 24h 跟進)
7	Special Summer Promotion	OTHER	⏭ Skipped (促銷信, 不回覆)
8	Question About Refund Policy	OTHER	⏭ Skipped (政策詢問, 非請求)

6 封皆成功寄出 (回傳各自的 Message ID), 全程無人工介入、無重複回覆。

2.3 Evaluation Summary — 對照 §2.6 (逐欄相符)

統計項目	Expected (§2.6)	實際結果	判定
Processed	8	8	✅
REFUND_REQUEST	2	2	✅
RETURN_REQUEST	2	2	✅
COMPLAINT	2	2	✅
OTHER	2	2	✅
Replies Sent	6	6	✅
Skipped	2	2	✅

```
Processed 8 emails
REFUND_REQUEST : 2
RETURN_REQUEST : 2
COMPLAINT      : 2
OTHER          : 2
Replies Sent   : 6
Skipped        : 2
```

2.4 Refund Agent — Success Criteria (§2.7)

Success Criteria	判定
100% 分類正確	✅ PASS (8/8 正確)
All actionable emails receive replies	✅ PASS (6/6 已回)
OTHER emails are skipped	✅ PASS (2/2 跳過)
Summary statistics match expected counts	✅ PASS (逐欄相符)

Success Criteria	判定
No duplicate responses	✅ PASS (每封僅一次回覆)

三、整體驗收標準 (對照 PDF 第 3 節)

#	Overall Acceptance Criteria	判定
1	Calendar Agent 正確執行查詢、排程、空檔搜尋	✅ PASS
2	Refund Agent 正確分類所有郵件	✅ PASS
3	Refund Agent 寄出適當回覆	✅ PASS
4	摘要報告與預期結果相符	✅ PASS
5	執行過程無 runtime error	✅ PASS

結論：系統通過全部驗收標準 (5/5) 。

附錄：Agent 原始摘要輸出

Refund Agent — Final Report (原文節錄)：

```
Found 8 unread emails matching customer service query.
- "Question About Refund Policy"      → OTHER      → Skipped
- "Return Request for Keyboard"        → RETURN_REQUEST → Replied
- "Return Request for Wireless Mouse" → RETURN_REQUEST → Replied
- "Refund Request for Order #1002"     → REFUND_REQUEST → Replied
- "Refund Request for Order #1001"     → REFUND_REQUEST → Replied
- "Poor Service Experience"            → COMPLAINT      → Replied
- "Very Disappointed"                  → COMPLAINT      → Replied
- "Special Summer Promotion"           → OTHER          → Skipped
6 threaded replies sent successfully.
```

完整逐步畫面 (含 ReAct 標記與彩色卡片) 可用 `python demo/view_terminal.py refund` 與 `python demo/view_terminal.py calendar` 重播；原始 JSON 存於 `demo/data/` 。